SANJAY SAHA

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About me

I am a Research Fellow at the Augmented Human Lab at the National University of Singapore (NUS), developing real-time Small Language Models (SLMs) and explainable Human-Robot Interaction (HRI) systems for safety-critical environments. Bringing over 4 years of combined research and industry experience, I previously built production-scale multimodal video understanding and real-time moderation pipelines as a Machine Learning Engineer at TikTok. I hold a PhD in Computer Science from NUS, with research spanning deepfakes, biometric cybersecurity, and synthetic media. Driven by a commitment to translating rigorous research into reliable AI systems, my work has been published in top venues including Pattern Recognition, IEEE Transactions on Biometrics, ICCV, and WACV. Alongside my research, I bring over four years of award-winning university teaching experience.

Research and Industry Experience

- **Augmented Human Lab, National University of Singapore [🌐]** — *May 2026 – Present*
  Post-doctoral Research Fellow — Singapore
  - Developing a context-aware, multimodal Small Language Model (SLM) interface to enable real-time, privacy-preserving decision support in high-risk scenarios.
  - Leading the iterative design and deployment of an explainable Human-Robot Interaction (HRI) system to improve operator task accuracy and reduce cognitive load during critical missions.
  - Managing a team of researchers and engineers while actively contributing to the formulation and submission of research grant proposals.

- **NexZ Pty. Ltd. [🌐]** — *Nov 2025 – Present*
  Founding AI Learning Engineer (Part-time) — Australia
  - Working on an Enterprise AI Engine for workflow automation, integrating RAG-based retrieval with vector databases to ground LLM responses in private organizational data.
  - Developing a hybrid MLLM framework supporting both local and remote models, enabling secure, multimodal reasoning for automated task orchestration.
  - Engineered scalable data pipelines for vector synchronization and semantic search, reducing manual process latency through autonomous AI agents.

- **TikTok Pte. Ltd. [🌐]** — *Jul 2024 – May 2026*
  Machine Learning Engineer — Singapore
  - Develop and maintain large-scale video deduplication services as part of the Content Intelligence team, supporting real-time content moderation on TikTok LIVE.
  - Design and deploy computer vision and multimodal learning models for duplicate and near-duplicate video detection across visual, audio, and textual signals.
  - Build scalable inference and retrieval pipelines to analyze high-volume livestream data, improving moderation accuracy and user experience.
  - Collaborate closely with product, infrastructure, and trust & safety teams to productionize research ideas into robust ML systems.

- **Department of Computer Science, National University of Singapore [🌐]** — *Jan 2024 – Jul 2024*
  Research Associate — Singapore
  - Conducted research under the iGyro (Information Gyroscope) project on detecting and assessing AI-generated and manipulated visual media.
  - Developed algorithms for analyzing deepfakes and AIGC content, contributing to research on trustworthy and responsible media analysis.
  - Supported dataset construction, experimental evaluation, and publication preparation for peer-reviewed venues.

- **Centre for Trusted Internet and Community, NUS [🌐]** — *May 2022 – Jan 2024*
  Research Assistant — Singapore

- Worked on the project *Deep and Cheap-Fakes – Effects on Audience's Attitudes, Knowledge, and Literacy*.
- Studied diverse deepfake generation and face synthesis techniques to create controlled experimental stimuli.
- Produced large-scale, curated deepfake datasets used in user studies to analyze the impact of manipulated media on audience perception and media literacy.
- Collaborated with interdisciplinary teams across computer science and social science research.

- **Invigilo Technologies [🌐]** *Jun 2020 – Apr 2022*
  Research Engineer (Part-time) Singapore
  - Developed computer vision solutions for automated safety violation detection in construction work sites.
  - Implemented state-of-the-art video analytics techniques including object detection, multi-object tracking, human action recognition, pose estimation, and proximity analysis.
  - Contributed to end-to-end deployment of vision-based monitoring systems in real-world industrial environments.
- **National University of Singapore [🌐]** *Aug 2018 – May 2022*
  Graduate Teaching Assistant (Part-time) Singapore
  - Conducted tutorial sessions for undergraduate and graduate modules including Discrete Mathematics, Database Systems, Biometric Authentication, Applied Machine Learning for Business Analytics, Advanced Analytics and ML, Visual Computing, and SCALE AI & ML.
  - Mentored students in assignments, projects, and exam preparation across multiple semesters.
  - Awarded a place on the Department's *Teaching Assistant Honors' List* in 2022.
- **ByteDance [🌐]** *May 2021 – Jul 2021*
  Research Intern (PhD) Singapore
  - Contributed to a TikTok video duplication detection project during PhD internship.
  - Worked with transformer-based model variants and image-matching algorithms for large-scale video similarity analysis.
  - Generated analytical reports and insights using SQL to support model evaluation and decision-making.

Teaching Experience

- **National University of Singapore (NUS)** *Aug 2018 – May 2022*
  Graduate Teaching Assistant (Part-time) Singapore
  - Responsibilities included preparing tutorial materials, conducting tutorial sessions, grading, exam preparation and occasionally designing assignments. I also provided one-to-one consultations after class to address students' questions and strengthen core understanding, for mostly introductory (Level-1) modules. Awarded a place on the Department's *Teaching Assistant Honors' List* in 2022.
    - test
    - Discrete Mathematics (CS1231S: AY2019-20-sem1, AY2020-21-sem1; TIC1201: AY2019-20-sem2).
    - Database Systems (CS2102: AY2018-19-sem2).
    - Biometric Authentication (CS5332: AY2021-22-sem2).
    - Applied Machine Learning for Business Analytics (BT5153: AY2020-21-sem2, AY2021-22-sem2).
    - Advanced Analytics and Machine Learning (BT5151: AY2021-22-sem2).
    - SCALE AI&ML (AY2021-22-sem1, AY2021-22-sem2, AY2022-23-sem2).
    - SoC Visual Computing Workshop (AY2021-22-sem1).
- **University of Asia Pacific (UAP)** *Oct 2017 – Jul 2018*
  Assistant Professor Dhaka, Bangladesh
  - Responsibilities included designing curricula, preparing course materials, and delivering core lectures for foundational computer science courses. I also provided dedicated mentorship to final-year students, guiding their undergraduate research projects to completion, which included leading one group to successfully achieve a peer-reviewed publication.
    - Structural Programming Language (2014 to 2018).
    - Algorithm Design (2015 to 2018).
    - Object Oriented Programming (2015 to 2017).
    - Guided students in Final Year research projects (2017 to 2018).

- **United International University (UIU)** *Jul 2014 – Sep 2017*
  Lecturer Dhaka, Bangladesh
  - Responsibilities included developing comprehensive course materials and conducting lectures across a wide range of foundational theoretical and programming modules. Alongside my daily teaching duties, I actively engaged in undergraduate academic mentoring and consistently contributed to the ongoing development of the department's curriculum.
    - Structural Programming Language (2014 to 2018).
    - Algorithm Design (2015 to 2018).
    - Object Oriented Programming (2015 to 2017).
    - Theory of Computation (2015 to 2017).
    - Discrete Mathematics (2014 to 2015).
    - Contributed to curriculum development and undergraduate academic mentoring.

EDUCATION

- **National University of Singapore** *Dec, 2024*
  PhD in Computer Science Singapore
  - GPA: 4.00/5.00
  - Thesis: Digital Face Synthesis: Safeguarding Against Threats (link)
- **University of Dhaka** *Feb, 2016*
  Master of Science in Computer Science Dhaka, Bangladesh
  - GPA: 3.54/4.00
- **University of Dhaka** *Mar, 2014*
  Bachelor of Science in Computer Science Dhaka, Bangladesh
  - GPA: 3.86/4.00

PROFESSIONAL MEMBERSHIPS

- **IEEE: Inst. of Electrical and Electronics Engineers** (97892456) *2021 – Present*
- **IEEE Biometrics Council**, Society Membership *2021 – Present*
- **IEEE Computational Intelligence Society**, Society Membership *2023 – Present*
- **AAAI: Association for the Advancement of AI** (648056) *2024 – Present*
- **AAAI Subgroups / Affiliates** AIIDE, EAAI, HCOMP, IAAI, ICAPS, KR *2024 – Present*
- **ACM: Association for Computing Machinery** (4895265) *2025 – Present*

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, SQL
- **Machine Learning:** PyTorch, Keras, other ML libraries
- **Database Systems:** HIVE, PostgreSQL, MySQL
- **Other Tools & Technologies:** AWS, Docker

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION • RANKINGS: CORE (CONF.), SJR QUARTILE (JOURNAL)

8 publications in SJR Q1-ranked journals and 2 in CORE A/A*-ranked conference venues.

[J.12] **Saha, Sanjay**; Hussain, Syed Ali; Soto-Sanfiel, María T.. (2026). **Nostalgic appeal in prosocial deepfakes of deceased and living artists: effects on emotional engagement and credibility**. *Behaviour & Information Technology*. [SJR: Q1]

[J.11] Fu, Gina Junhan; Soto-Sanfiel, María T.; **Saha, Sanjay**. (2026). **Resurrecting Historical Figures: The Persuasive Power of Deepfakes versus Text-Based First-Person Narratives**. *Mass Communication and Society*. [SJR: Q1]

[C.11] Yew, Wei Chee; Xu, Hailun; **Saha, Sanjay**; Fan, Xiaotian; Ong, Hiok Hian; Wang, David Yuchen; Sarkar, Kanchan; Yang, Zhenheng; Guan, Danhui. (2025). **Dynamic Content Moderation in Livestreams: Combining Supervised Classification with MLLM-Boosted Similarity Matching**. Manuscript (arXiv preprint: arXiv:2512.03553).

[J.10] Chen, Ken; Seneviratne, Sachith; Wang, Wei; Hu, Dongting; **Saha, Sanjay**; Hasan, Md Tarek; Rasnayaka, Sanka; Malepathirana, Tamasha; Gong, Mingming; Halgamuge, Saman. (2025). **AniFaceDiff: Animating stylized avatars via parametric conditioned diffusion models**. *Pattern Recognition*, pp. 112017. [SJR: Q1]

[J.9] Stragapede, Giuseppe; Vera-Rodriguez, Ruben; Tolosana, Ruben; Morales, Aythami; DeAndres-Tame, Ivan; Damer, Naser; **Saha, Sanjay**; et al. (2025). **KVC-onGoing: Keystroke Verification Challenge**. *Pattern Recognition*, Vol. 161, pp. 111287. [SJR: Q1]

[J.8] Soto-Sanfiel, María T.; Angulo-Brunet, Ariadna; **Saha, Sanjay**. (2025). **Deepfakes as narratives: Psychological processes explaining their reception**. *Computers in Human Behavior*, Vol. 165, pp. 108518. [SJR: Q1]

[J.7] Soto-Sanfiel, María T.; Angulo-Brunet, Ariadna; **Saha, Sanjay**. (2025). **Motivations for (not) sharing deepfakes on social media**. *The Communication Review*, pp. 1–26. [SJR: Q2]

[S.4] Bahavan, Nadarasar; **Saha, Sanjay**; Chen, Ken; Seneviratne, Sachith; Rasnayaka, Sanka; Halgamuge, Saman. (2025). **Unmasking the Unknown: Facial Deepfake Detection in the Open-Set Paradigm**. Manuscript (arXiv preprint: arXiv:2503.08055).

[C.10] Kung, Han-Wei; Varanka, Tuomas; **Saha, Sanjay**; Sim, Terence; Sebe, Nicu. (2025). **Face anonymization made simple**. In *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 1040–1050. IEEE. [CORE: A]

[J.6] Nguyen, Kim-Ngan; Rasnayaka, Sanka; Wickramanayake, Sandareka; Meedeniya, Dulani; **Saha, Sanjay**; Sim, Terence. (2024). **Spatio-temporal dual-attention transformer for time-series behavioral biometrics**. *IEEE Transactions on Biometrics, Behavior, and Identity Science*, Vol. 6(4), pp. 591–601. [SJR: Q1]

[C.9] Soto-Sanfiel, María T.; **Saha, Sanjay**. (2024). **Predicting deepfake enjoyment: A machine learning perspective**. In *International Conference on Human-Computer Interaction*, pp. 384–402. Springer. [CORE: ?]

[J.5] **Saha, Sanjay**. (2024). **Digital Face Synthesis: Safeguarding Against Threats**. PhD Thesis, *National University of Singapore*.

[C.8] **Saha, Sanjay**; Perera, Rashindrie; Seneviratne, Sachith; Malepathirana, Tamasha; Rasnayaka, Sanka; Geethika, Deshani; Sim, Terence; Halgamuge, Saman. (2023). **Undercover Deepfakes: Detecting Fake Segments in Videos**. In *IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*, pp. 415–425. IEEE. [CORE: A* (main conf.)]

[C.7] Stragapede, Giuseppe; Vera-Rodriguez, Ruben; Tolosana, Ruben; Morales, Aythami; DeAndres-Tame, Ivan; Damer, Naser; Fierrez, Julian; Garcia, Javier-Ortega; Gonzalez, Nahuel; Shadrikov, Andrei; et al. (2023). **IEEE BigData 2023 keystroke verification challenge (KVC)**. In *2023 IEEE International Conference on Big Data (BigData)*, pp. 6092–6100. IEEE. [CORE: B]

[C.6] **Saha, Sanjay**; Sim, Terence. (2020). **Is Face Recognition Safe from Realizable Attacks?**. In *2020 IEEE International Joint Conference on Biometrics (IJCB)*, pp. 1–8. IEEE. [CORE: B]

[C.5] Rasnayaka, Sanka; **Saha, Sanjay**; Sim, Terence. (2019). **Making the most of what you have! Profiling biometric authentication on mobile devices**. In *2019 International Conference on Biometrics (ICB)*, pp. 1–7. IEEE. [CORE: B]

[C.*] Ahmed, Shahjalal; Islam, Md; Hassan, Jahid; Ahmed, Minhaz Uddin; Ferdosi, Bilkis Jamal; **Saha, Sanjay**; Shopon, Md. (2019). **Hand Sign to Bangla Speech: A Deep Learning in Vision based system for Recognizing Hand Sign Digits and Generating Bangla Speech**. In *SUSMCOM 2019*. [CORE: not ranked]

[J.4] Rahman, M Saifur; Rahman, Md Khaledur; **Saha, Sanjay**; Kaykobad, M; Rahman, M Sohel. (2019). **Antigenic: an improved prediction model of protective antigens**. *Artificial Intelligence in Medicine*, Vol. 94, pp. 28–41. [SJR: Q1]

[J.3] Rahman, M Saifur; Shatabda, Swakkhar; **Saha, Sanjay**; Kaykobad, Mohammad; Rahman, M Sohel. (2018). **DPP-PseAAC: a DNA-binding protein prediction model using Chou's general PseAAC**. *Journal of Theoretical Biology*, Vol. 452, pp. 22–34. [SJR: Q2]

[J.2] Islam, Md Mofijul; **Saha, Sanjay**; Rahman, Md Mahmudur; Shatabda, Swakkhar; Farid, Dewan Md; Dehzangi, Abdollah. (2018). **iProtGly-SS: Identifying protein glycation sites using sequence and structure based features**. *Proteins*, Vol. 86(7), pp. 777–789. [SJR: Q1]

[C.4] Ali, Muhammad; Taniza, Farzana Afrin; Niloy, Arefeen Rahman; **Saha, Sanjay**; Shatabda, Swakkhar. (2018). **Prediction of bacteriophage protein locations using deep neural networks**. In *IEMIS 2018*, pp. 29–38. Springer. [CORE: not ranked]

[J.1] Shatabda, Swakkhar; **Saha, Sanjay**; Sharma, Alok; Dehzangi, Abdollah. (2017). **iPHLoc-ES: identification of bacteriophage protein locations using evolutionary and structural features**. *Journal of Theoretical Biology*, Vol. 435, pp. 229–237. [SJR: Q2]

[C.3] Ahmed, Al Amin Neaz; Haque, HM Fazlul; Rahman, Abdur; Ashraf, Md Susam; **Saha, Sanjay**; Shatabda, Swakkhar. (2017). **A participatory sensing framework for environment pollution monitoring and management**. Manuscript (arXiv preprint: arXiv:1701.06429).

[C.2] **Saha, Sanjay**; Elahi, Md. Mamun; Islam, Md. Mahfuzul; Ahmed, Shabbir. (2017). **An efficient successive authentication scheme for vehicular ad-hoc networks**. In *2017 20th International Conference of Computer and Information Technology (ICCIT)*. [CORE: not ranked]

[C.1] Md Shakil Hossain; Sk. Shariful Islam Arafat; S M Al-Hossain Imam; Md. Mahmudul Hasan; Md. Mofijul Islam; **Saha, Sanjay**; Shatabda, Swakkhar; Tamanna Islam Juthi. (2017). **VIM: A Big Data Analytics Tool for Data Visualization and Knowledge Mining**. In *2017 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE 2017)*. [CORE: not ranked]